

Depth Discovery in a Market with Expandable Limit Orders:
An Investigation of the U.S. Treasury Market

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Abstract

This paper studies the brokered interdealer trading activity of primary dealers in the secondary market for U.S. Treasury securities. Limit orders in this market are *expandable*: specifically, quotes represent a firm-price commitment to trade but also serve as an invitation for brokered one-on-one quantity negotiation. As a result, we find that few transactions (0.6% of dollar volume) receive price improvement but many transactions (45.1% of dollar volume) discover greater-than-quoted depth. We find evidence consistent with dealers' using expandable limit order strategies to reduce costs associated with information leakage and stale limit orders. Expandable orders, however, create queues for other dealers who must wait to transact. We find evidence consistent with dealers' willingness to use another broker at worse prices rather than wait in the queue of a broker with quantity negotiations in progress.

How deep is a market? In the study of market microstructure, *depth* describes the amount that can be bought at the ask and sold at the bid. It is often used as a measure of liquidity.¹ Many studies examine *price discovery*, which is the process by which prices converge to the full-information value of an asset.² Far less research investigates *depth discovery*, which we define as the process by which traders determine the true depth that can be traded at a particular price.³ Depth discovery is less than straightforward in many markets. Some market structures, such as open-outcry futures exchanges, do not lend themselves well to the aggregation of quoted depth. In other markets (e.g., the NYSE, Nasdaq market, and Tokyo Stock Exchange), some participants can observe more information about depth than others can. In addition, markets may have rules that allow traders to conceal the extent of their willingness to trade, such as provisions for hidden orders and crossing. Some markets make depth at prices outside the best bid and ask opaque to all participants. In many markets, depth is unobservable ex ante because transactions receive better than quoted prices.⁴

Our investigation of depth discovery is timely for several reasons. First, recent empirical findings underscore the complexities of depth discovery. Sofianos and Werner (1997) find that floor traders provide substantial depth beyond the specialist quoted depth on the NYSE. They conclude that, because floor traders “work” customers’ orders, specialist quoted depth understates true depth. Bacidore, Battalio, and Jennings (2000) find that 16 percent of NYSE system market orders are larger than the quoted depth and that 70 percent of these orders receive “depth improvement”, i.e., they are executed in full at the quoted price. Handa, Schwartz, and Tiwari (1999) find that 5 percent of total system orders receive depth improvement on the AMEX. Barclay, Christie, Harris, Kandel, and Schultz (1999) state that even following the Securities and Exchange Commission 1997 rule changes, quotes posted by Nasdaq dealers do not necessarily represent the extent of their willingness to trade. Nasdaq dealers sometimes trade in excess of quoted depth. Quoted depth may also overstate true depth as market makers “may not always honor their quotes”. Reiss and Werner (1996) indicate that on the London Stock Exchange, dealers post quotes but negotiate trades by phone. They find that dealers give better prices on larger trades. Biais, Hillion, and Spatt (1995) find that on the Paris Bourse, where market orders are automatically converted to limit orders when they exceed quoted

¹ Other liquidity measures include width of the bid-ask spread, immediacy, and resiliency. For a discussion of the dimensions of market liquidity, see Harris (1990).

² Theoretical models of price discovery include Leach and Madhavan (1992, 1993). Recent empirical studies of price discovery include Cao, Ghysels, and Hatheway (2000) and Huang (2000).

³ An important exception is research on block trading, in which traders or brokers search for counterparties for large trades [e.g., Burdett and O’Hara (1987) and Madhavan and Cheng (1997)].

⁴ Blume and Goldstein (1992), Lee (1993), and Petersen and Fialkowski (1994), among others, provide evidence that trades often receive better prices than best displayed bid and offers, i.e., “price improvement”, for NYSE-listed securities. See Handa, Schwartz, and Tiwari (1999) for AMEX and Reiss and Werner (1996) for the London Stock Exchange. Huang and Stoll (1996) compare price improvement for NYSE and Nasdaq stocks.

depth, converted market orders attract liquidity. Biais et al. suggest this finding indicates the existence of undisclosed liquidity: specifically, traders monitor the book and place orders after observing market order conversion. Finally, competition among electronic communication networks (ECNs) to provide markets without designated market makers suggests market designs that allow low-cost depth discovery are of practical interest.

This paper provides an empirical investigation of depth discovery in the brokered interdealer market for U.S. Treasury securities. Quotes in this market are particularly interesting in that they represent a firm-price commitment to trade but also serve as an invitation for one-on-one quantity negotiation. As a result, few transactions (0.6% of dollar volume) occur at prices within the quoted spread but many transactions (45.1% of dollar volume) exceed quoted depth.

The brokered interdealer Treasury market consists of a small number of dealers⁵ who use a small number of brokers⁶ to meet anonymously to negotiate transactions.⁷ Through a broker, dealers can post anonymous quotes (i.e., limit orders) and can act on the quotes of others. In assigning trading priority, a broker extends to each of the two dealers for whom he has most recently brokered a trade the right-of-first-refusal on additional volume.⁸ The right-of-refusal protocol, in essence, creates a market in which order size is *expandable*: a dealer whose order has just been executed maintains trading priority as long as he is willing to trade again at that price. Because order size is expandable, a dealer can display an anonymous *minimum-size* limit order to attract another dealer willing to transact at a fixed price. Once the two dealers are matched on price, they communicate quantity demands through the broker to discover depth quickly (i.e., they expand the size of their orders). To the extent that *expandable orders* allow dealers to influence how and when information about their willingness to trade is revealed, they can reduce the costs of information free-riding and stale limit orders. Since brokers are agents for dealers but do not trade for their own account, broker revenues consist of

⁵ For example, the Federal Reserve Bank of New York reports that for third quarter 1998, 14 primary dealers accounted for 76.2% of the Treasury-bill and 81.6% of the Treasury-coupon brokered interdealer volume.

⁶ The interdealer broker firms in this market are Cantor Fitzgerald, Garban, Hilliard Farber, Liberty, Intercapital (formerly RMJ), and Tullett and Tokyo.

⁷ Most trades between primary dealers (i.e., “interdealer trades”) in this market are brokered [Fleming (1997)]. Strict anonymity is maintained even after the transactions are completed. Netting, title transfer, and payment are handled through the Government Securities Clearing Corporation. Dealers are under no obligation to trade, and brokers generate revenue only when dealers trade.

⁸ Although this right-of-refusal feature has been used in the brokered interdealer Treasury market for more than 20 years [Garbade (1978)], it is rarely mentioned in the literature. Exceptions are Garbade (1978), Stigum (1990), U.S. General Accounting Office (1990), Umlauf (1991), and Fleming (1997).

volume-based commissions.⁹ Consequently, both dealers and brokers in the U.S. Treasury market may garner substantial benefit from the use of an expandable order protocol.¹⁰

While an expandable order protocol may offer substantial benefits, it introduces a potentially important friction, first recognized in the literature by Garbade (1978). Queues may develop as other dealers wait their turns to trade. We hypothesize that impatient dealers, instead of queuing, go to other brokers (when they exist) for faster execution. Competition among brokers, therefore, may be an important feature in this market: it allows dealers to use another broker when the expected cost of waiting in a broker's queue is too high. As a result, while the use of an *expandable order protocol* in a multiple-broker market may encourage dealers to submit limit orders and discover depth quickly, *competition among brokers* may cap the benefit that dealers gain from paired communication.

Analyzing a unique data set of matched workups, quotes, and transactions, we find that dealers often expand their orders to discover depth in this market. We find trading patterns consistent with the use of expandable order strategies to lessen costs associated with information leakage and stale limit orders. Dealers appear to use expandable order strategies more frequently during periods of greater asymmetric information, such as during hours outside the New York business day and for off-the-run issues. There is no evidence that for on-the-run trading during New York business hours, dealers can interpret asymmetric depth of the limit order book as a valuable information signal and attempt to free-ride on it. Dealers' flexibility to use expandable order strategies mitigates but does not eliminate the potential for free-riding, however. We find that for off-the-run and overseas trading, asymmetric depth may provide a valuable signal about the sign of the next trade. We find evidence consistent with dealers' willingness to use another broker, sometimes at worse prices, rather than wait in the queue of a broker who has negotiations underway. The results of an ordered probit analysis suggest that dealers expect to obtain better prices by executing through an alternative broker than by waiting in a queue as liquidity is exhausted and quotes are revised.

To our knowledge, we provide the first empirical investigation of the use and relevance of an expandable order trading protocol. In doing so, we extend the market microstructure literature of the secondary U.S. Treasury market. Fleming (1997), Fleming and Remolona (1997), Elton and Green (1998), Fleming and Remolona (1999), and Balduzzi, Elton, and Green (2001) use transaction level data to analyze activity in the brokered interdealer Treasury market but do not use data at the right-of-refusal level.

⁹ The exception is Cantor Fitzgerald, which trades for its own account. Cantor Fitzgerald offers brokerage services to a broad list of clients while the other 5 firms broker mainly for primary dealers.

¹⁰ We will use *right-of-first-refusal protocol* and *expandable order protocol* interchangeably throughout this paper.

Recent research cites the importance of this Treasury market as one motivation for modeling interdealer trading.¹¹ However, the expandable order protocol used in the Treasury market permits dealers more flexible trading strategies than those analyzed in existing models. As the opportunity to use expandable orders is central to the selection of Treasury trading strategies, inferences drawn from models omitting such opportunities may be incorrect. In some cases, the actual Treasury trading protocol's net incentive may be the exact opposite of that characterized. For example, in Vayanos' model (1999) without expandable interdealer orders, dealers trade slowly because order size signals the size of private endowment shocks and the level of future trading activity. Expandable orders may have been introduced precisely because of the slow trading incentive Vayanos identifies. We conjecture that brokers continue to use the expandable order protocol, in part, because it encourages dealers to trade quickly.¹²

We provide an important distinction between "hidden limit orders" and what we term "expandable limit orders". Biais, Hillion, and Spatt (1995) and Panchapagesan (1998) describe hidden limit orders as the electronically *staged* limit orders of the Paris Bourse ("CAC") and Toronto Stock Exchange ("CATS"), respectively. Other types of hidden limit orders have been defined for the NYSE. McNish and Wood (1995) define hidden limit orders as the limit orders specialists have chosen not to display. Sofianos and Werner (1997) describe how NYSE floor brokers sometimes "work" large limit orders: the brokers ask the specialist to display only part of the order. Like these orders on the CAC, CATS, and NYSE, part of the expanded order in the Treasury market represents demand that has been "hidden" from the market. But the hidden component is different from that of CAC and CATS, where traders pre-commit to the demand that will be revealed in stages, and on the NYSE, where traders fully communicate their willingness to trade to the specialist or floor broker. During bilateral negotiations in the Treasury market, a dealer's true desire to trade can expand or contract based on negotiations and market conditions. The only component of demand that is communicated through brokers is the actual willingness (or lack thereof) to trade the next incremental quantity. Both hidden limit orders and expandable orders could be termed "iceberg orders" as what is exposed may only be the tip of a larger volume.¹³ We believe, however, it is worth noting that in the

¹¹ Goodhart and O'Hara (1997) call for theoretical and empirical work to fill the void in the fixed income market microstructure literature. Vayanos (1999) and Viswanathan and Wang (1999) include the Treasury market in the list of interdealer markets that motivate their analyses. Theoretical models of markets with brokered interdealer trading are also presented in Ho and Stoll (1983), Saporta (1997), Vogler (1997), Naik, Neuberger, and Viswanathan (1999), Lyons (1996, 1997), Cao and Lyons (1999), and Vayanos (2000).

¹² In contrast with the Vayanos (1999) model, Vayanos (2000) incorporates noise in an interdealer trading model and shows that dealers trade quickly.

¹³ We thank an anonymous referee for this insight.

Treasury market, no one knows the portion that will remain underwater, not even the counterparties who place the initial limit order: trading is dynamic, mutable, and time consistent.

Finally, our findings contribute to a discussion started by Garbade (1978) as to whether brokered or direct interdealer trading is more efficient at eliminating price arbitrage. We conclude that competition among brokers is an important feature in markets that allow expandable orders. Queue-jumping is a solution to the problem created when brokered bilateral negotiations bottleneck trading. In the absence of the opportunity to jump, the demand for trade is censored and the market, in some sense, fails or at least performs inefficiently (like other markets where allocations are determined by non-price considerations). An interesting aspect of the design of the Treasury market is that while the expandable order protocol, which allows sequential paired bargaining on depth, has advantages, it must be combined with a multiplicity of brokers or intermittent batch-crossing sessions to limit the negative externality imposed on those waiting to trade. Attempts to use the expandable order protocol as the *only* market-clearing mechanism in *non-competitive* broker markets or electronic exchanges may not result in durable market structures.

The organization of the remainder of the paper is as follows. Section I details the expandable order clearing protocol. Section II describes data and summary statistics. Section III provides the empirical investigation of depth discovery using expandable orders. Section IV investigates the effects of broker competition when a market permits expandable orders. Section V concludes.

I. Description of Expandable Order Protocol

Most interdealer brokers in the Treasury market use a protocol consisting of the following steps:

Step 1.) *Limit order* placement, followed by

Step 2.) *Responding order* placement (and agreement to pay the broker a transaction fee), followed by

Step 3.) A negotiating period, during which the paired dealers are offered right-of-refusal options to transact additional volume at the initial volume price.¹⁴

In the limit-order step, a dealer places an order with a broker to buy (or sell) a particular quantity at a particular price. In the responding-order step, another dealer instructs the same broker that he will take the opposite side of the transaction at the limit-order price. The responding-order

¹⁴ The possible exception is Cantor Fitzgerald. Several dealers told us in December 1998 that Cantor Fitzgerald no longer offers these right-of-refusal options in the cash market. However, in the futures market, Cantor Financial Futures Exchange, Inc., includes the right-of-refusal feature in their market-clearing protocol under their “Exclusive Time” provision [Cantor Financial Futures Exchange, Inc., By-Laws and Rules, NY12533.105106.1].

dealer may tell the broker he is interested in transacting some of, all of, or more than the limit-order quantity. Dealers can cancel their limit orders unless they have been matched with a responding order. Minimum order size is \$1 million. Buy limit orders are called *bids*. Sell limit orders are *offers* or *asks*. A responding order is a *hit* if it is a request to sell at the price shown in the limit-order bid. A *take* is a responding order requesting to buy at the limit-order offer price. Thus, dealers “hit the bid” or “take the offer” of other dealers.

Each broker firm installs a proprietary screen in dealers’ trading rooms and displays its highest-priced bid (*best bid*) and lowest-priced offer (*best offer*). Dealers see not only each broker firm’s best bid and best offer from these screens, but also the aggregate quoted depth at each broker firm’s best bid and offer. Typically, dealers have the screens of all broker firms in their trading rooms.

With the possible exception of Cantor Fitzgerald, brokers do not trade for their own account. Broker revenues are volume-based commissions. Brokers make a clear distinction between limit orders and responding orders because only responding orders pay commissions. As a result, a broker displaying a limit order from Dealer A to buy at price Y and a limit order from Dealer B to sell at price Y may show a bid and ask at price Y and yet not match the orders. Alternatively, Dealer A and Dealer B may have their limit orders at price Y with different brokers. Dealers and brokers refer to the market as *locked* as long as no transaction takes place while the spread between the best bid and best offer is zero.

The third step in the market-clearing protocol is the quantity negotiation period, or *workup process*. During the workup process, the broker attempts to obtain agreement between the limit-order and responding-order dealers to transact a higher volume than initially agreed. The broker grants each dealer a right-of-refusal option to transact additional quantity at the just-agreed price, conditional on the other party’s agreement to trade more. A dealer who exercises the option to trade another increment continues to receive a right-of-refusal option to trade again at that price, until the counterparty interest is exhausted. As a limit-order or responding-order dealer completes his trading, the broker brings into the workup process the next dealer, if one exists, who has queued up with an order to trade at this same price. If one counterparty completes his trading, the *other* counterparty continues to retain a right-of-refusal option until *his* trading is completed.

To inform other dealers that volume negotiations are in progress, the broker firm immediately displays the initially matched quantity and each subsequent increment in flashing fields on its screen. Each volume increment (including the initial quantity) is called a *workup*. When all dealer interest is exhausted at the initially-agreed price on at least one side of trading, the broker firm posts the transaction to its screen by clearing the workup field and adding the total of the workups to its screen’s aggregate daily transaction volume field.

Orchestrating trade requires that brokers assign trading priorities to incoming orders. The most aggressively priced limit order bid and offer are displayed on the broker's screen. If a broker firm receives best-price limit orders from more than one dealer, it displays the aggregate of the limit order depth on its screen. The dealer who first places a responding order to hit or take a displayed limit order enters the workup process initially. The dealer who first placed the limit order at that price is given priority over all other limit orders, *even when the posted depth is from multiple dealers*.

Queues form as dealers wait behind those with the right-of-refusal options. Dealers cannot be certain of their position within a broker's queue. As explained by the U.S. General Accounting Office (1990), different employees within a broker firm represent different dealers. Each employee's compensation is an increasing function of the volume he brokers. As a result, employees within a broker firm sometimes argue over the queue positions of dealers they represent.

Figure 1 shows an example of the market-clearing protocol simplified to show a market consisting of only 2 brokers and 4 dealers. The example includes the information each broker would display on its proprietary screen.¹⁵

The example in Figure 1 illustrates that the posted transaction volume (\$4 M) is the total accumulation of 3 individual workups at a price of $100 \frac{1}{64}\%$ of face value. Broker X considers the transaction complete and posts this volume to the screen's transaction field only after exhausting willingness to trade on at least one side of the market at the $100 \frac{1}{64}$ price. Dealer C, who submits his limit order later than Dealer A, starts a queue behind Dealer A. Because Dealer A does not exhaust Dealer D's demand for trade, Broker X can bring Dealer C into the volume negotiations with Dealer D. The transaction posted by Broker X, therefore, represents pairwise negotiations of Dealer A and Dealer D as well as Dealer C and Dealer D.

As shown in Figure 1, the size of the *first* workup (\$1 M) is not greater than Dealer A's initial limit order. Even though Dealer D's responding order (\$5 M) is for a larger amount, Dealer D waits while Dealer A considers exercising the option to trade more. Because Broker X's screen displays the aggregate depth of Dealer A and Dealer C's limit orders, the *initial* workup of \$1 M is for less than Broker X's displayed depth of \$2 M.

¹⁵ The final column of Figure 1 shows GovPX screen information, which is explained in Section II.

II. Data and Summary Statistics

Responding to Congressional pressure to increase transparency in the Treasury markets, the primary dealers and most of the interdealer brokers, in 1990, formed a joint venture called GovPX to capture quote and transaction data electronically and to provide it on a real-time basis to corporations, institutional investors, and other subscribers. GovPX screens show subscribers the consolidation of contributing brokers' best bids and offers, workups, and transactions. Our data set consists of all GovPX transaction and quote data from the brokered interdealer secondary market for 5-year U.S. Treasury notes for October 1-29, 1997.

The final column of Figure 1 shows the information the GovPX screen would display. Consistent with institutional characteristics described in Section I and with GovPX terminology, we define a *transaction* as the posting by an individual broker of a volume increment in the *transaction field* of his firm's proprietary screen. It can consist of one or more workups (i.e., a *single-workup transaction* or *multiple-workup transaction*, respectively). All workups in multiple-workup transactions have the same price and sign. The example of Figure 1 shows only one transaction: a multiple-workup hit transaction of \$ 4 M at a price of $100 \frac{1}{64}$. The sizes of the first, second, and third workups in the transaction are \$ 1 M, \$ 2 M, and \$ 1 M, respectively.

The five GovPX-member brokers who provide real-time feeds from their proprietary screens to the GovPX electronic system are Garban, Hilliard Farber, Intercapital (formerly RMJ), Liberty, and Tullett & Tokyo. One of the large interdealer brokers, Cantor Fitzgerald, is not involved in the joint venture or reporting. Using data from the Federal Reserve Bank of New York, we estimate GovPX-member firms broker 71% of the volume of Treasury coupon securities with maturity of 5 years or less that is traded between primary dealers. The remaining interdealer volume is brokered by Cantor Fitzgerald or traded directly (i.e., without brokers).

In addition to providing electronically-collected quote, workup, and transaction data to subscribers on a real-time basis, GovPX markets a historical data set of these time-stamped quote, workup, and transaction records. The GovPX historical data are time-stamped observations of limit orders aggregated across the brokers (price and size), signed workups (price and size), and signed transaction postings (price and size). GovPX creates a new time-stamped observation whenever any variable (excluding time) changes. The GovPX data do not reveal the identities of the dealers or brokers on any quotes or trades.

The GovPX *bidprice* (*askprice*) variable is the price of the highest-priced bid limit order (lowest-priced ask limit order) posted by the 5 brokers, if any exists. The GovPX *bidsize* (*asksize*) variable is the sum across all broker inputs of limit orders at that price. We observe only what GovPX

displays, which is the inside spread. Thus, we observe less than dealers observe, which is the highest-priced bid limit order and lowest-priced ask limit order individually from each broker's screen.

Unlike many microstructure data sets, the GovPX data set includes *signs* (*hit* or *take*) for all workups and posted transactions as input by the brokers. A posted transaction's sign is defined as the sign of the responding order, consistent with interdealer broker conventions described in Section I.

The GovPX historical data list each time-stamped workup individually, consistent with the broker terminology for inputting workups on display screens. Although GovPX does not indicate to which posted transaction a workup belongs, the unique nature of the expandable order protocol and the manner in which GovPX captures the raw data allow us to partition the raw data into clusters, greatly simplifying the job of matching workups with transactions. In fact, 82.5% of clusters have only one possible way of matching workups with transactions. For the remainder, we adopted two approaches: one maximizes the number of workups associated with a given transaction and one minimizes the number. These approaches give rise to almost indistinguishable results. The results presented here are those from the approach that minimizes the number of workups associated with a given transaction and, therefore, biases against finding evidence of expanded orders. An appendix showing samples of raw data, clusters, and a variety of examples of how we match workups with transactions is available from the authors upon request.

Table I provides summary statistics for transactions, quotes, and workups for the three most recently auctioned 5-year issues. Market participants refer to the most recently auctioned issue as the *on-the-run* issue. The rest of the issues are called *off-the-run* issues. During the sample period, only one 5-year issue, Cusip 9128273J9 ("J9"), was on-the-run. This issue was designated on-the-run from September 25-October 29, 1997. Prior to that, Cusip 9128273G5 ("G5") was the on-the-run issue from August 28-September 24, 1997. Cusip 9128273C4 ("C4") was on-the-run from July 24-August 27, 1997. Table I shows that, in comparison with the on-the-run issue, the two off-the-run issues rarely traded.

Average transaction size is \$ 8.3 million for the on-the-run issue and \$ 4.4 million and \$ 6.1 million for the other issues. Ninety percent of transactions are for \$ 20 million or less. Although we do not have data on primary dealer trades with customers (e.g., regional banks, insurance companies, and fund managers), Chakravarty and Sarkar (1999) analyze Treasury security trades reported by insurance companies for 1995 through 1997 and conclude the average transaction size approximately equals the average size of interdealer Treasury trades brokered by GovPX members.

Both bid and ask limit orders are present for the on-the-run issue 89% of the 24-hour trading day. Many of the absences of a two-sided spread occur between 5:00 p.m. and 6:00 p.m. New York time when trader activity shifts geographically from New York to Tokyo. Two-sided spreads are

absent more of the time for the off-the-run issues. Conditional on the presence of both bid and ask, the time-weighted average spread for the on-the-run issue is narrower than for either of the off-the-run issues. To calculate the time-weighted average spread conditional on the existence of both bid and ask, we calculate the best ask less best bid in effect for every second both are present, sum these differences, and then divide by the number of seconds when both bid and ask are present, as in Blume and Goldstein (1997).

On average, for the on-the-run issue, every 18 seconds there is a change in quote prices or sizes, causing an updated GovPX record. The off-the-run issues average less frequent quote revision but substantially higher ratio of quote revisions to number of transactions.

Table I shows there are an average of 2 workups for every transaction. For the on-the-run issue, the average time between first workup display and transaction volume posting is 13 seconds for single-workup transactions and increases for multiple-workup transactions.

Figure 2 shows daily closing prices for the 3 issues. Consistent with the GovPX “trading day” designation, a day’s closing price is the price of the last transaction recorded before 6:00 p.m. that day (and after 6:00 p.m. the previous day). While the on-the-run issue (J9) traded every day, the most-recently-designated off-the-run issue (G5) did not trade on October 27. The second-most-recently designated off-the-run issue (C4) traded only on 10 of the 20 trading days. The largest single-day price changes occurred on:

- October 8, when Federal Reserve Chairman Alan Greenspan’s remarks on employment suggested interest rate tightening,
- October 27, when the NYSE first halted trading under Rule 80B following a 550-point drop in the Dow Jones Industrial Average (DJIA), and
- October 28, as the DJIA rebounded 337 points from the previous day.

Clearly, the activity in the U.S. equity market on October 27 and 28, 1997, was extraordinary. Daily volumes (number of transactions and dollar volume) are shown for our on-the-run Treasury issue in Figure 3. The two days with the highest trading volume are these two days. We considered the possibility that dealers used different trading strategies on these two days. To test whether this possibility affects our conclusions, we performed a robustness check of our empirical findings by repeating our analysis for the data set with October 27 and 28 excluded from the sample. Excluding these two days does not alter our conclusions.

Figure 4 shows price changes between transactions for the on-the-run issue for the entire period. Almost sixty percent (59.9%) of on-the-run transactions were posted with no change in price from the previous transaction. Another 25.7% of transactions were posted with a price change from the previous transaction of a positive or negative $\frac{1}{128}$ of one percent of face value.

III. Depth Discovery with Expandable Orders

A. Evidence that Dealers Expand Orders

First, we look for evidence that dealers expand orders. If dealers expand orders, transactions will consist of multiple workups. We find that multiple-workup transactions account for 38.9% of the on-the-run trades and for 58.8% of the on-the-run dollar volume. For the off-the-run issues, they make up 39.6% of trades and 62.8% of dollar volume. Of the multiple-workup transactions, 58% of have just two workups, 18% have three workups, and 6% of multiple-workup transactions have more than 6 workups.

We note that observations of multiple-workup transactions are necessary but not sufficient evidence of expanded orders. As described in Section I, such observations could indicate the sequential participation of multiple limit order dealers or multiple responding order dealers in each transaction. A stronger indication of expandable order usage is the observation of multiple-workup transactions that exceed quoted depth. If transaction size exceeds quoted depth, dealers have negotiated quantity incremental to their limit orders (i.e., they “expanded” their limit orders).

Table II shows transactions by their size and depth quoted immediately prior to first workup. In the spirit of Fleming (1997), we define:

- 1) *New York hours* as 7:30 a.m. to 6:00 p.m., and
- 2) *Non-New York hours* as 6:00 p.m. to 7:30 a.m.

Panel A shows that 25.8% of on-the-run transactions during New York hours are multiple-workup transactions of size that exceeds quoted depth. These transactions make up 45.3% of the dollar volume.¹⁶

We check how our estimate is affected by our inability to observe the depth of quotes other than at the best bid and ask of any broker. Not having to rely on the GovPX screen, dealers can observe each broker firm’s best bid and offer on individual broker screens. A dealer may hit the bid or take the offer displayed by a broker firm that is inferior to that displayed by another broker firm. When this occurs, we observe transaction price, volume, and workups, but not the depth that the dealer observed quoted at that price. Only a tiny fraction of trades (0.6% of dollar volume) have *positive* price improvement, i.e., “price improvement” in the conventional sense that the trade occurs *within* the inside spread. We define a transaction as having *negative* price improvement if it has an initial hit

¹⁶ Due to constraints imposed by the GovPX data, this approach may underestimate the extent dealers expand orders because we observe only depth aggregated across brokers and aggregated across dealer limit orders. We are also unable to discern *new* traders if they arrive with limit orders *during* the workup process.

workup price lower than the GovPX best bid or an initial take workup price higher than the GovPX best ask. These transactions are included in the amounts shown in the body of Table II and presented in isolation in the “Memo” column. They account for 6.9% of all on-the-run transactions and are 6.4% of the dollar volume. After excluding trades with negative price improvement, multiple-workup trades with size exceeding quoted depth account for 45.1% of all on-the-run dollar volume.¹⁷

Panel B of Table II shows that multiple-workup trades of size greater than quoted depth account for a higher proportion of trading during non-New York trading hours. They also occur as a higher fraction for the off-the-run issues, as shown in Panel C.

It is interesting to contrast our findings with what we would expect to observe with a pure limit order book protocol.¹⁸ In a pure limit order book market, we would not expect a single transaction’s realized depth to exceed quoted depth. Instead, a large market order could quickly run through a series of limit orders at a variety of prices, creating a series of transactions at various prices. In contrast, in an expandable order protocol, while paired negotiations are underway, the broker firm can no longer update its screen with a “quoted depth” of the limit order dealer who holds the right-of-refusal option: the broker does not know this “depth” until negotiations are complete. If limit order dealers exercise right-of-refusal options to increase transaction volume, depth is *discovered* sequentially by responding order dealers without its being quoted first to all dealers.

B. *Why Do Dealers Use Expandable Order Strategies?*

We next investigate whether dealers use expandable order strategies to lessen the costs associated with information leakage (i.e., the *free-riding* problem) and stale limit orders (i.e., the *free-option* problem). There is always a potential for information free-riding when market participants interpret a large order as a signal of valuable private information and seek to free-ride on the signal.¹⁹ Various authors have suggested that free-rider problems for large traders could be mitigated by splitting orders across time, across brokers, or across exchanges. In the Treasury market, large orders may signal private

¹⁷ Table II shows that for some transactions, the size of the *first* workup exceeds the quoted depth. Some of these observations have negative price improvement. Regarding the others, one explanation is that brokers sometimes may accumulate negotiated quantity increments prior to posting the first workup to the screen, as discussed in Section II. Another explanation may be errors related to identifying workups with transactions other than for the 82.5% of data clusters that have unique identification.

¹⁸ We define a *pure* limit order protocol as one that enforces strict price, time priority and distinguish it from a hybrid that allows *hidden* limit orders.

¹⁹ Burdett and O’Hara (1987) discuss information free-riding as a potential problem for block trading in the upstairs market. Fishman and Longstaff (1992) analyze the free-riding problem that traders face when they use brokers to trade in futures markets. Pagano and Roell (1993) describe the free-riding problem in the context of traders whose orders are front-run by their stockbrokers.

information about customer order flow and about future trades.²⁰ Other dealers, observing large quoted depth, may attempt to free-ride on the information signal by attempting to trade ahead of the large quoted depth. If placing large limit orders encourages free-riding by others, dealers will attempt to conceal the extent of their willingness to trade by placing small initial limit orders and then working up volume in paired negotiations. As a result, large transactions will tend to consist of more than a single workup. Furthermore, once paired negotiations are underway, a dealer divulges the extent of his willingness to trade to only one other dealer. Therefore, the potential for free-riding is reduced during the workup process.²¹ As a result, dealers may be more willing to trade larger increments during paired negotiations than in an initial limit order.

In contrast to the free-riding problem, in which a dealer must consider whether others will trade ahead of his order, a dealer must also consider the free-option problem, in which his order is executed to his regret.²² In the Treasury market, dealers continually update their private information sets by talking and trading with customers. As a result, limit orders can become stale quickly as dealers update their own willingness to trade. By placing a small initial limit order and working volume up in increments, a dealer reduces the value his initial order provides to those who may react more quickly than he can. Upon having his initial limit order executed, the dealer can update his information set and decide whether to trade again. Once paired in the workup process, a dealer commits his willingness to trade for a short and finite time period, thus providing a less valuable

²⁰ Madigan and Stehm (1994) report that dealers indicate an important reason for maintaining anonymity in the brokered Treasury market is to protect proprietary speculative positions and inventories. Based on discussions with primary dealers, Meulendyke (1998) reports dealers trade “to achieve inventories consistent with customer demands and with interest rate expectations”. Meulendyke indicates dealers also pursue “arbitrage” transactions in the sense of Dow (1998), trying to “capitalize on the price differences between cash markets and futures and options markets.” Speculative positions may also take the form of attempts to corner or “squeeze” a particular Treasury issue [Department of Treasury et al. (1992), Jordan and Jordan (1996), *Wall Street Journal* (2000)]. Duffie (1996) analyzes “repo specialness” that can result from dealer-orchestrated or inadvertent squeezes.

²¹ The potential for free-riding is reduced but not eliminated during the workup process. The counterparty in the paired negotiation, upon hearing the size of the next proposed increment, can decline that increment and reverse direction.

²² Copeland and Galai (1983) introduce the free-option problem associated with limit orders as a partial explanation for positive bid-ask spreads. Dealers quote prices that balance expected losses to informed traders, who see the quote as an in-the-money option, with expected gains from uninformed traders, who demand immediacy. Rock (1991) applies the free-option characterization to limit order traders competing with a market maker who, after observing order flow information, decides how much to participate in the quotes. Stoll (1992) provides examples of the free-option problem where limit orders are “picked off” across exchanges [by floor traders using the Intermarket Trading System (ITS)] and across dealers [by Small Order Execution System (SOES) traders]. Stoll argues that continuous market monitoring alleviates the free-option problem but that market structures that allow market orders to run quickly through the limit order book, such as automated exchanges, will exacerbate the problem. Stoll further argues that, because a large limit order provides a large free option, traders wanting to transact large quantities [institutional traders] will avoid placing limit orders, possibly choosing to “hire someone to ‘work’ the order”. O’Hara (1995) discusses the free-option characterization as applied to markets without designated market makers and reasons that designing electronic

option than when he places an initial limit order. So, as is the case when dealers use expandable order strategies to reduce free-riding, dealers, concerned about the cost of free options of their initial limit orders, may be more willing to trade larger increments during paired negotiations.

In summary, if dealers use expandable order strategies to reduce the costs of free options and information free-riding, large transactions will tend to consist of multiple workups and the size of the initial workup will tend to be smaller than the average size of the rest of the workups in transactions.²³ We first analyze whether dealers tend to use multiple-workup transactions when they execute large trades. We then examine whether dealers tend to trade larger size increments during paired negotiations than they trade as initial orders.

To examine whether large transactions are more likely to consist of more than one workup, we partition transactions by their size into the following three categories:

- 1) *small trades*, which are transactions of \$ 1 M to \$ 4 M face value,
- 2) *medium trades*, which are transactions of \$ 5 M to \$ 9 M face value, and
- 3) *large trades*, which are transactions of \$ 10 M or greater.

By choosing these size ranges, approximately a third of on-the-run trades are partitioned into each of the three size categories. We use multiple-workup transactions as the proxy for expanded orders and single-workup transactions as the proxy for orders that are not expanded.

Panel A of Table III shows multiple- and single-workup transactions as a function of trade size for the on-the-run issue. Multiple-workup transactions are a higher proportion of trades as trade size increases. Large trades are more likely to consist of multiple workups than of a single workup. Using χ^2 tests, we test the null hypothesis that trade size does not affect the probability of occurrence of multiple- and single-workup transactions. The null is easily rejected at the 1% significance level.

Panel B of Table III shows a similar analysis for the off-the-run issues. As for the on-the-run issue, the relative frequency of multiple-workup transactions increases with trade size. Large trades are more likely to consist of more than one workup than of a single workup. As for the on-the-run issue, the null that trade size does not affect whether transactions have more than one workup is easily rejected at the 1% significance level.

Next, for multiple-workup transactions, we examine how initial workup size compares with the average size of the rest of the workups in the transaction. Panel A of Table IV reports that for about half of the on-the-run transactions, the size of the initial workup is smaller than the average size

limit-order books may be difficult: “few traders submit large limit orders, preferring instead to work any intended trade via a series of small market orders”.

²³ In comparison, in a pure limit order book, a large transaction might be more likely than a small transaction to execute against multiple orders. But the *initial* order that it executes against at a given price would not necessarily tend to be smaller than the *next* orders it executes against at that same price.

of the rest of the transaction's workups. As shown in Panel B of Table IV, transactions with initial workups that are smaller than the average of the rest of the workups make up a higher proportion of the off-the-run trades.

In summary, large transactions tend to consist of multiple workups. For multiple-workup transactions, the average size of workups during paired negotiations is more likely to be larger than the size of the initial order. These findings are consistent with dealers' use of expandable order strategies to mitigate free-riding and free-option problems.

C. Use of Expandable Order Strategies in Periods of Information Asymmetry

In the previous section, we concluded that there is evidence consistent with dealers' use of expandable order strategies to reduce the costs of free options and information free-riding. In this section, we test whether dealers use these strategies to a greater extent during periods of greater information asymmetry. Limit orders provide more valuable options in periods of greater information asymmetry. Additionally, in these periods, large initial limit orders may be interpreted as stronger signals of private information and may be more likely to invite unwanted free-riding in the form of competition from other dealers. Other dealers may be more likely to attempt to achieve execution over an existing limit order by placing a limit order at the same quote through another broker, by placing a limit order at a more aggressive quote, or by placing a responding order against a limit order on the opposite side of the market. To reduce this unwanted competition from others, dealers should use expandable strategies to a greater extent during periods of greater information asymmetry. Specifically, during these periods:

- 1) large transactions should be more likely to consist of multiple workups,
- 2) the first workup of transactions should tend to be smaller, and
- 3) the size of the initial workup should be more likely to be smaller than the average size of the rest of the workups in transactions.

Previous researchers [Madigan and Stehm (1994), Fleming (1997), Elton and Green (1998), Green (1999), Balduzzi, Elton, and Green (2001)] identify the following as possible regimes of greater information asymmetry in the brokered interdealer Treasury market as the result of some dealers' being able to update or react to new information more quickly than others:

- 1) Trading during hours outside the New York business day.
- 2) Trading in off-the-run issues.
- 3) Trading up to fifteen minutes *following* macroeconomic announcements.

Madigan and Stehm (1994) and Fleming (1997) indicate that trading during the New York business day is structurally different from trading conducted outside the New York hours. Madigan and Stehm indicate there are fewer analysts, traders, and brokers working during non-New York hours. These structural differences suggest there may be opportunities for some dealers to obtain and react more quickly to valuable information than others can. For example, Madigan and Stehm report that a higher fraction of trades are done directly between dealers and therefore not necessarily disseminated through the broker network. Fleming (1997) concludes price changes during overseas trading hours are unbiased predictors of transaction prices when trading resumes in New York the following day. He compares prices measured at the end of the Tokyo and London trading days, however, and does not analyze the potential for information asymmetry within the Tokyo and London trading days.

Elton and Green (1998) show off-the-run issues are subject to pricing errors relative to on-the-run issues as the result of liquidity and tax-timing effects. Off-the-run issues trade infrequently in the brokered market, and customer-dealer trading is less transparent than the brokered market. As a result, dealers who trade off-the-run issues more frequently may be able to update or react to new information obtained from the customer-dealer market more quickly than other dealers can.

Balduzzi, Elton, and Green (2001) conclude that there is information asymmetry up to fifteen minutes following macroeconomic announcements. Using GovPX quote and transaction data, but not workup data, they find:

- In general, prices in the Treasury market adjust to news within the first minute following scheduled macroeconomic announcements.
- Trading volume and volatility are greater for up to an hour following these announcements.
- Bid-ask spreads are wider for the first minute following announcements but narrow to normal levels within 5 to 15 minutes following the announcement.

They suggest the short period of wider spreads indicates the presence of informed traders: dealers “may have differing ability to process” the just-released information. Once the bid-ask spread returns to normal levels, elevated trading and volatility result from uninformed liquidity trading.

Our tests of whether dealers are more likely to use expandable order strategies during regimes of greater information asymmetry are joint tests. We are jointly testing 1) the dealers’ tendency to use expandable order strategies and 2) whether the trading regimes we select are actually perceived by dealers to have more asymmetric information than the regimes with which we compare them.

First, we examine whether *large transactions* are more likely to consist of multiple workups during periods of greater information asymmetry. The first regime of greater information asymmetry that we examine is on-the-run trading during non-New York hours. As shown in Panel C of Table III, large transactions are more likely to have multiple workups during non-New York trading hours. While 60.6% of large New York transactions have more than one workup, 81.0% of non-New York transactions have more than one workup. We use population proportion t-tests of the null hypothesis that the proportions of transactions that have more than one workup, conditional on large trade size, are equal for New York and non-New York trading regimes. We reject the null hypothesis at the 1% significance level. We also apply 3-way χ^2 tests and reject the null hypothesis of randomness conditional on each transaction size and on each trading regime at the 1% significance level.

The next regime of greater information asymmetry that we investigate is off-the-run trading. Comparing Panels A and B of Table III, we observe that large trades are also more likely to consist of multiple workups for off-the-run issues than for the on-the-run issue. We reject the null hypothesis that the proportions of transactions with more than one workup, conditional on large trade size, are equal for off-the-run and on-the-run issues at the 1% significance level. Using 3-way χ^2 tests, we reject the null hypothesis of randomness conditional on each transaction size and on each trading regime at the 2% significance level.

The last regime of greater information asymmetry that we investigate is that following macroeconomic announcements. Scheduled macroeconomic announcements occur at 8:30 a.m. for half the trading days in our sample. These announcements are Weekly Initial Jobless Claims, Consumer Price Index, Durable Goods Orders, Housing Starts, Producer Price Index, Retail Sales, Unemployment Report for September, and Trade Balances. To control for other possible time-of-day effects, we compare trading activity during the 8:30 a.m. to 8:45 a.m. time period on days that have announcements with days that do not. Consistent with Balduzzi et al. (2001), we find that:

- Trading volume and volatility are higher for an hour following the announcement than for that same hour on non-announcement days, and
- Bid-ask spreads widen in the fifteen minutes following the announcements and then return to non-announcement day levels.²⁴

We have only one 9:15 a.m. announcement and only three 10:00 a.m. announcements in our sample, so we do not test these time periods.

²⁴ Green (1999) uses a structural model of intraday price formation to decompose the effective bid-ask spread into adverse selection and transaction costs components. Green concludes information asymmetry significantly

As reported in Panel D of Table III, we find that large trades are more likely to consist of more than one workup during the fifteen minutes following an announcement. Using population proportion t-tests, we cannot reject, however, the null hypothesis that the proportions of transactions with more than one workup, conditional on large trade size, are equal during the fifteen minutes following macroeconomic announcements and during the same time period on days when an announcement is not made (p-value = 0.17).

We next examine whether initial workups tend to be smaller during periods of greater information asymmetry. The smallest workup size allowed by brokers is \$ 1 million. This is also the smallest limit order any dealer is allowed to place. Table V shows that, for the on-the-run issue, transactions with a minimum-size initial workup account for a much higher percentage of transactions during non-New York trading hours than during New York trading hours. Table V also shows that transactions with a minimum-size initial workup account for a higher percentage of transactions for the off-the-run than for the on-the-run issues.

For the on-the-run issue, Table V shows that use of minimum-size initial workups is higher than during the fifteen minutes following macroeconomic announcements than during that same time period for days when no announcement is made. The p-value for the χ^2 test of randomness is 0.10, however.

Next, we examine the means and medians of the initial workups for each of the above trading regimes. As shown in Table VI, means and medians are smaller for off-the-run issues than for the on-the-run issue. For the on-the-run issue, means and medians are smaller for the initial workups of trades made outside the New York business day than those made during New York business hours. In the fifteen minutes following macroeconomic announcements, medians are slightly smaller and means are not significantly different.

Lastly, for multiple-workup transactions, we test whether initial workup size is more likely to be smaller than the average size of the rest of the workups in the transaction when there is greater information asymmetry. Panel C of Table IV shows initial workup size is more likely to be smaller than the rest of the workups during non-New York hours than during New York hours. Initial workups are also more likely to be smaller for off-the-run issues than for the on-the-run issue. Panel D of Table IV shows that initial workups are more likely to be smaller than the rest of the workups during the period following macroeconomic announcements, but we cannot reject the null of randomness when information asymmetry is measured in this manner (p-value = 0.19).

increases in the *half-hour* following announcements. We do not find evidence that information effects extend to this longer timeframe.

In summary, the evidence is consistent with dealers' using expandable order strategies relatively more frequently during periods of greater information asymmetry, such as hours outside the New York business day and for off-the-run issues. We find weak evidence that dealers are more likely to use expandable order strategies during the fifteen minutes following a macroeconomic announcement.

D. Free-Riding and Asymmetric Depth

We investigate whether trade sign is related to asymmetric thickness of the bid and ask immediately prior to the trade. If dealers use expandable order strategies to conceal the extent of their willingness to trade until they are paired in the workup process, depth displayed on broker screens will not forecast the full quantity that informed limit order dealers are willing to trade at the quoted price. Uncertainty about true depth lessens the information content of limit order asymmetry. In fact, the option to expand limit orders could eliminate the potential for information free-riding by eliminating any relationship between limit order depth and the direction of future trades.

Testing whether trade direction and asymmetric thickness of the limit order depth are uncorrelated is more than just a test of the potential for information free-riding. It is also a test of whether thickness on one side of the limit order book "crowds out" the placement of additional limit orders on that side of the book, as suggested by Parlour (1998). Parlour models a transparent limit order book with small orders that are not expandable and with time priority strictly enforced as a rationing rule. Parlour shows that, even in markets without asymmetric information, systematic patterns in trade direction emerge. Traders know that new limit orders placed on the thicker side of the limit order book have a lower probability of execution. As a result, all else equal, traders are more likely to place market sell (buy) orders than limit sell (buy) orders when the book is thicker at the ask (bid). Therefore, in an open limit order book market with price and time priority, hit (take) transactions will be more likely than take (hit) transactions when the book is thicker at the ask (bid). In comparison, in an expandable limit order market, because true depth is not known by any participant, we would expect "crowding out" to be less important and asymmetric thickness to be less correlated with future trade direction.

We partition the relative thickness of the aggregate limit order book into the following three groups:

- 1) depth at the best ask greater than depth at the best bid,
- 2) depth at the best ask equal to depth at the best bid, and
- 3) depth at the best ask less than depth at the best bid.

If dealers interpret asymmetric depth as a valuable information signal and attempt to trade on that information, hits (takes) will be more likely than takes (hits) when the limit order book is thicker at the ask (bid).

For all on-the-run trades, 48.5% are hit transactions and 51.5% are take transactions. Panel A of Table VII shows that when ask depth exceeds bid depth, the probability of observing a hit increases to 49.6% versus 48.5% unconditionally. Take probability increases to 52.4% when bid depth exceeds ask depth versus 51.5% unconditionally. We run a χ^2 test of the null hypothesis that the probability of observing a hit or take is not correlated with the relative thickness of the book. The p-value is significant (0.02); but as described below, partitioning on-the-run trading into New York and non-New York hours is more informative.

Results for the off-the-run issues are shown in Panel B of Table VII. Hit probability increases to 56.8% when ask depth exceeds bid depth versus 50.1% unconditionally. Take probability increases from 49.9% unconditionally to 59.3% when bid depth exceeds ask depth. We reject the null hypothesis of randomness (p-value = 0.03).

Panel C of Table VII shows that by partitioning on-the-run trading into New York and non-New York hours, we find some evidence that trade direction and relative thickness are correlated during non-New York hours (p-value < 0.01). However, for New York trading hours, we cannot reject the null hypothesis of trade sign randomness (p-value = 0.16). At conventional confidence levels, we also cannot reject the null hypothesis of trade sign randomness for the fifteen minutes following macroeconomic announcements.

In summary, for the on-the-run issue during New York hours, it does not appear that dealers can interpret asymmetric limit order depth as a valuable information signal and attempt to free-ride on it. Likewise, there is no evidence that thickness on one side of the limit order book “crowds out” the placement of additional limit orders on that side of the book for on-the-run trading during New York hours. In contrast, for on-the-run trading during non-New York business hours and for the off-the-run issues, there is evidence consistent with potential information or crowding out effects during times of limit order asymmetry.

IV. What Are the Effects of Broker Competition in an Expandable Order Market?

An important feature of the brokered interdealer Treasury market is that each of several competing brokers offers limit order placement. Three implications of broker competition in an expandable order market are:

- *If and when competition among brokers limits the value of paired negotiations, we will observe time-overlapping transactions.*
- *If and when an expandable order protocol constrains immediacy or causes price pressure, we will observe time-overlapping transactions of the same sign across brokers at different prices.*
- *If competition among brokers using the expandable order protocol results in a well-functioning market, we should not observe crossed spreads for any significant length of time.*

In most decentralized markets where multiple orders can be worked at any given time, we would also expect to observe time-overlapping transactions.²⁵ Unlike in other decentralized markets, however, Treasury brokers offer the right-of-refusal option to paired dealers, and this potentially bottlenecks the trading of other dealers. The right-of-refusal feature may cause impatient dealers who are not first in the queue to cancel orders at the broker firm with workups in progress and to submit orders at a competing broker firm. In addition, a dealer who has not yet placed an order can observe workups in progress and decide to trade. Instead of queuing behind other dealers at the broker with trading in progress, he can try to execute through another broker. In either of these cases, the dealer is attempting to *jump the queue* of the broker who has workups in progress. To the extent that time-overlapping transactions at a second broker make trading volume unavailable to either of the parties initially paired, competition among brokers has limited the protection of the paired negotiation.

In some markets, traders may choose to use particular brokers (those who provide preferential or “personal” service to them) over others.²⁶ Three important elements of the brokered interdealer Treasury market are: 1) competition among broker firms, 2) competition among employees at each broker firm, and 3) dealers’ desires for anonymity. Dealers have the screens of all broker firms in their trading rooms. In a real sense (through the installation of all the screens), the dealers have created important execution options. As the quotes, workup increments, and completed transactions of each broker are immediately transparent, each dealer quickly knows whether he is getting best execution. Additionally, each broker has multiple employees, each looking out for the interests of his designated dealer. Competition among the broker firm’s employees helps to assure that time priority at the broker firm is maintained. Consequently, the opportunity to provide special treatment (in return for order flow) is limited, and it seems unlikely that “personal service” or reputation for special treatment can survive in this market. Finally, one reason that dealers usually use brokers (instead of trading direct) is to preserve anonymity: dealers want to protect private order flow information about

²⁵ We thank the editor for raising this point.

²⁶ We also thank the editor for raising this point.

their inventory positions. It seems doubtful, therefore, that any dealer would use only a single broker firm as doing so would jeopardize anonymity. Although our data do not permit us to identify individual brokers or dealers, dealers have told us that the ability to use different brokers at different times is an important regularly exercised option.

One option for dealers who do not want to wait in a queue is to respond to limit orders displayed through another broker, possibly at worse prices, resulting in transactions with negative price improvement. Dealers can also attempt to jump the queue by placing limit buy (sell) orders at higher (lower) prices than take (hit) workups in progress. Such aggressive order strategies introduce the possibility that the best bid will exceed the best offer (i.e., a *crossed* spread). If the brokered market for interdealer trading is typically arbitrage-free, crossed spreads should be quickly eliminated. For high-volume dealers, commissions for responding orders are about \$10 per million-dollar face value of the transaction. This is about one-eighth of the minimum tick size. As a result, even the tightest crossed spread provides an arbitrage opportunity. We believe crossed spreads usually involve multiple brokers. We assume individual broker firms, valuing repeat business, have incentives to alert dealers who are inadvertently attempting to submit arbitrageable limit orders.

First, we look for evidence of crossed spreads. We find little evidence that crossed spreads are sustainable. Best bid exceeds the best ask less than 1% of trading hours. For off-the-run issues, crossed spread observations usually coincide with very large negative spreads of about minus one percent of face, which may indicate broker data errors.

We estimate *attempts* at queue jumping in the form of aggressive limit order placement. We refer to a limit order as aggressive if it is submitted while another transaction is in progress and at a disadvantageous price relative to the transaction in progress. We calculate that 2.4% of all take transactions and 3.0% of all hit transactions experience at least one aggressive limit order.

Next, we analyze time-overlapping transactions. We define a transaction as overlapping if its initial workup occurs after the initial workup of another transaction but before the posting of that transaction's final volume. For example, suppose an initial workup is posted for Transaction #1 and several seconds later, but prior to the posting of Transaction #1's final volume, Transaction #2's initial workup is posted. Transaction #2 is the overlapping transaction.

Table VIII shows transaction overlap for on-the-run trading.²⁷ Almost 40% of transactions overlap at least one other transaction. A tiny fraction of trades (only 0.6% of dollar volume) have positive price improvement. About 5% of all transactions are overlapping transactions that are the result of responding orders at worse than best quotes (i.e., negative price improvement). Comparison

²⁷ We find no evidence of overlapping transactions for either off-the-run issue.

of overlapping and non-overlapping transactions indicates that a higher fraction of the overlapping transactions execute with negative price improvement. Only 2.8% of non-overlapping transactions have negative price improvement versus 13.7% for overlapping transactions. Table VIII also shows negative price improvement is not simply due to responding order dealers who demand larger transactions. Transactions with negative price improvement are not significantly larger.

We find that purchases tend to follow purchases and sales tend to follow sales. As Biais et al. (1995) note in their study of the centralized limit order book of the Paris Bourse, this finding is consistent with order splitting, informed traders' repeated trading, uninformed traders' imitating the trading of informed traders, and similar reactions to new information. We find that 84.3% of *overlapping* trades have the same sign as the trade they overlap.

To investigate further the decision to use an overlapping trade, we model price changes between transactions as a function of dealers' observables: quote size and the previous transaction's price change, sign, dollar volume, number of workups, and state of completion. We use an ordered probit model because most of the observed price changes are accounted for using only 5 values, as shown for the on-the-run issue in Figure 4.

Using notation consistent with Hausman, Lo, and MacKinlay (1992), we denote the observed price change for trade k as Z_k and model the probability of observing a specific price change s_i as

$$P(Z_k = s_i | X_k, W_k) = \begin{cases} P(X'_k \beta + \varepsilon_k \leq \alpha_1 | X_k, W_k) & \text{if } i = 1, \\ P(\alpha_{i-1} < X'_k \beta + \varepsilon_k \leq \alpha_i | X_k, W_k) & \text{if } 1 < i < m, \\ P(\alpha_{m-1} < X'_k \beta + \varepsilon_k | X_k, W_k) & \text{if } i = m, \end{cases}$$

$$= \begin{cases} \Phi\left(\frac{\alpha_1 - X'_k \beta}{\sigma_k}\right) & \text{if } i = 1, \\ \Phi\left(\frac{\alpha_i - X'_k \beta}{\sigma_k}\right) - \Phi\left(\frac{\alpha_{i-1} - X'_k \beta}{\sigma_k}\right) & \text{if } 1 < i < m, \\ 1 - \Phi\left(\frac{\alpha_{m-1} - X'_k \beta}{\sigma_k}\right) & \text{if } i = m, \end{cases}$$

where

- Z_k $\equiv$ the price of trade k less the price of trade $_{k-1}$ (as percent of face value),
 $trade_{k-1}$ $\equiv$ the non- k trade for which a workup was most recently posted to the GovPX screen,
 X_k $\equiv$ variables that influence the unobservable continuous variable Z_k^* , which maps Z_k to s_i ,
 ε_k $\equiv$ the random disturbance associated with Z_k^* , assumed to be independently but not identically distributed $N(0, \sigma_k^2)$,
 W_k $\equiv$ variables that influence the conditional variance σ_k^2 ,
 α $\equiv$ the vector of partition boundaries to be estimated via maximum likelihood,
 β $\equiv$ the vector of coefficients to be estimated via maximum likelihood,
 m $\equiv$ the number of ordered observed price changes (i.e., $s_i \in [s_1, \dots, s_m]$),
 $\Phi(\bullet)$ $\equiv$ the standard normal cumulative distribution function.

We specify $X_k' \beta$ as

$$\begin{aligned}
 X_k' \beta = & \beta_1 Bid_k + \beta_2 Ask_k \\
 & + [1-IND_k] [\beta_3 Z_{k-1} + \beta_4 IBS_{k-1} + \beta_5 (V_{k-1} \bullet IBS_{k-1}) + \beta_6 (N_{k-1} \bullet IBS_{k-1})] \\
 & + [IND_k] [\beta_7 Z_{k-1} + \beta_8 IBS_{k-1} + \beta_9 (V_{k-1} \bullet IBS_{k-1}) + \beta_{10} (N_{k-1} \bullet IBS_{k-1})],
 \end{aligned}$$

where

- Bid_k $\equiv$ size (M\$) of best bid immediately prior to the first workup of trade k .
 Ask_k $\equiv$ size (M\$) of best ask immediately prior to the first workup of trade k ,
 IND_k $\equiv$ $\begin{cases} 1 & \text{if there is at least one other trade in progress when trade } k \text{ begins.} \\ 0 & \text{otherwise.} \end{cases}$
 Z_{k-1} $\equiv$ the price of trade $_{k-1}$ less the price of trade $_{k-2}$ (as percent of face value).
 IBS_{k-1} $\equiv$ $\begin{cases} +1 & \text{if trade }_{k-1} \text{ is a } take. \\ -1 & \text{if trade }_{k-1} \text{ is a } hit. \end{cases}$
 V_{k-1} $\equiv$ volume (M\$) of trade $_{k-1}$, calculated as the sum of its workups completed as of the first workup of trade k .
 N_{k-1} $\equiv$ number of workups in trade $_{k-1}$ as of the first workup of trade k .

We specify the conditional variance of ε_k as

$$\sigma_k^2 = 1 + \gamma^2 \Delta t_k,$$

where Δt_k is the time (in seconds) between the first workup of trade k and the last workup of trade $k-1$, and γ is a parameter estimated via maximum likelihood.²⁸

As assumed in models by Vayanos (1999) and Viswanathan and Wang (1999), we assume many of the trades in the brokered interdealer Treasury market are motivated by inventory hedges or other private order flow information.²⁹ We assume that these motivations predetermine the trade sign of dealers who place responding orders. We therefore estimate the ordered probit specification conditionally on whether the current trade is a hit or take. Because we also want to analyze the expected price change for dealers who consider placing limit orders, we also estimate the model not conditioned on the sign of the contemporaneous trade.³⁰

Given the difference in number of brokers, traders, and information analysts operating in U.S. and non-U.S. geographic locations [Madigan and Stehm (1994), Fleming (1997)], we assume trading in different geographic regions could be structurally different. We therefore estimate models separately for on-the-run transactions completed during New York and non-New York trading hours.

Maximum likelihood estimates for all the model specifications are reported in Table IX. Using likelihood ratio tests, we test the null hypothesis that whether the previous trade is still in progress does not matter. Specifically, we test $\beta_3 = \beta_7$, $\beta_4 = \beta_8$, $\beta_5 = \beta_9$, and $\beta_6 = \beta_{10}$. We reject the null for each model specification in Table IX at the 1% significance level. To test for serial correlation as the result of inclusion of too few lags of price changes, we follow Hausman et al., as derived by Gourieroux, Monfort, and Trognon (1985), and compute generalized residuals $\{\hat{\varepsilon}_k\}$, generalized fitted price changes $\{\hat{Z}_k\}$, sample correlation coefficients for $\hat{\varepsilon}_k$ with the j^{th} lag of generalized fitted price change $\{\hat{Z}_{k-j}\}$, and score test statistics to test the null hypothesis of no serial correlation due to omitted lagged price changes. We conclude the models are reasonably well specified.

As noted in Greene (1997), signs of coefficient estimates from ordered probit models can be used to interpret marginal effects of regressors on the probabilities of price changes only for the

²⁸ We use a simple dummy variable to analyze overlapping transactions. A logical extension would be to categorize the different types of simultaneous transactions.

²⁹ The crux of Ho and Stoll's (1983) canonical model of interdealer trading is that dealers trade with each other to share the inventory risk they incur from their customer-dealer trades. Using data from the London Stock Exchange, Reiss and Werner (1998) and Hansch, Naik, and Viswanathan (1998) construct inventory positions for dealers and conclude that inventory risk-sharing motivates interdealer trading in London Stock Exchange securities. Due to the anonymity of trading in the Treasury market, we cannot construct dealer inventories.

³⁰ Hausman et al. (1992) assume price change and trade direction for NYSE trading are endogenously determined and, as a result, do not estimate a model conditioned on contemporaneous trade sign.

extreme price changes (i.e., probability of price change equals plus or minus $3/128^{\text{th}}$ of 1% of face value). To interpret the implications of the models, we must analyze the expected value of price changes conditional on specific estimates for the regressors. Specifically, we construct cases that enable us to analyze the transaction price a dealer expects as a function of whether he trades using an overlapping or non-overlapping transaction.

First, we consider the case of a dealer who wants to *buy* on-the-run Treasury notes during New York hours. Using the fitted ordered probit model, we analyze the dealer's expected price, relative to the price of the previous transaction, conditional on whether the dealer executes a limit or responding order; the sign, price change, number of workups, and dollar volume of the previous transaction; and whether the previous transaction is still in progress. If the previous transaction is still in progress, the dealer's transaction will be an overlapping transaction. Specifically, the previous transaction is still in progress if at least one workup has been posted to the broker's screen's workup field but not to the transaction field.

Panel A of Table X shows the dealer's expected price relative to the previous transaction, conditional on executing via a responding take order, which we denote $E[Z_k | \text{take}]$. First, consider the scenario in which the previous trade is a take, with a price change (Z_{k-1}) of zero, one workup ($N_{k-1} = 1$), and a volume (V_{k-1}) of \$ 1 M. If this previous trade is still in progress, $E[Z_k | \text{take}]$ is \$19.96 per M\$ face (about 25% of $1/128^{\text{th}}$ of 1% of face value). However, if the previous transaction has been posted, $E[Z_k | \text{take}]$ is \$42.26 per M\$ face. In other words, all else equal, the dealer's expected price change is about twice as high if he does not execute his responding take order as an *overlapping* transaction. This predicted result is consistent with expectations that limit order dealers will revise ask quotes upward following the posting of take transactions.

Next, we investigate the impact of increasing dollar volume and number of workups for the previous transaction. Note that increasing the previous transaction's number of workups to 2 and volume to \$ 10 M has little impact on $E[Z_k | \text{take}]$. Additional analysis (not shown) indicates the effects of quote size and Δt_k are also small. However, the previous trade's price change (Z_{k-1}) does matter. Panel A of Table X shows that when the previous transaction's price change is $1/128^{\text{th}}$ of 1% of face value, $E[Z_k | \text{take}]$ is only - \$ 1.46 and - \$ 0.90 for the overlapping scenarios and \$ 19.00 for the non-overlapping scenarios. Comparing expected prices as a function of Z_{k-1} , we conclude that expected prices exhibit an important mean reversion component. For a responding take order, a dealer expects little upward price revision if the previous trade had a price increase. If the previous transaction had a price decrease, the dealer expects a greater upward price revision.

Next, we examine expected price changes if the dealer buys using a limit order. Panel B of Table X shows $E[Z_k | \text{hit}]$, which is the dealer's expected price change conditional on executing a limit buy order. $E[Z_k | \text{hit}]$ is the expected price change conditional on a trade that results from a responding *hit* order. The dealer's limit buy order executes only if another dealer places a responding *hit* order. In each scenario, the dealer expects to pay about \$60 per M\$ face less with a limit buy order than with a responding take order strategy. This difference represents an expected spread of almost $1/128^{\text{th}}$ of 1% of face. If his limit order executes, he also avoids broker commissions.

Panel C of Table X shows the expected price change not conditioned on trade sign, which we denote $E[Z_k]$. If the previous transaction is still in progress and the previous price change is zero, then $E[Z_k]$ is about \$10 per M\$ face, which is approximately equal to the commission a responding dealer pays the broker. $E[Z_k]$ doubles if the previous transaction has been posted.

The analysis of the scenarios in Table X suggests that any dealer who wants to *sell* notes should wait until the take transaction already in progress has been posted. In other words, the dealer should expect upward quote revision following completion of take transactions.

In analysis not shown, we alter each scenario to examine a dealer who is a seller when the trade in progress is a hit. Briefly, the same arguments apply as in Table X. The dealer, who wants to trade in the same direction as the previous transaction, is better off placing his order while the previous transaction is still in progress. Dealers expect quotes will be revised downward once the hit transaction in progress has been posted. Dealers who want to place a *take* responding order should wait to submit orders until the hit transaction has been posted.

Also in analysis not shown, we examine the implications of the models that were estimated using non-New York trading hours data. As in New York hours, dealers who want to trade in the same direction as the trade in progress expect a better price by executing an overlapping transaction than by trading after the transaction is completed by others. However, the mean reversion component, observed in results for New York hours, is smaller in the non-New York model conditioned on take transactions and not significant for the model conditioned on hit transactions and for the unconditional model. As in the New York hour model, we analyzed dollar volume and number of workups for trade_{k-1} , quotes sizes, and Δt_k , and found they have little or no impact on expected prices.

The results of the ordered probit analysis of price changes suggest dealers should attempt to trade via an overlapping transaction only if they want to trade in the same direction as the trade already in progress. If dealers expect price changes consistent with our analysis, we should observe that overlapping transactions tend to have the same sign as the transaction they overlap. This is consistent with our findings reported earlier that 84% of trades overlap trades of the same sign.

As discussed at the beginning of this section, we would expect time-overlapping transactions in decentralized markets where multiple orders can be worked at any given time. Therefore, we cannot conclude that all time-overlapping transactions (almost 40% of trades) represent queue-jumping. Quotes, workups, and transactions *across all brokers* are extremely transparent in this market, however. As a result, at a minimum, we interpret time-overlapping transactions with negative price improvement (13.7% of overlapping trades) as some evidence of dealers' decisions to jump the queue. We also see this result as showing that the costly option to leave the queue and trade through a competing broker is sometimes in the money, validating the need for competing brokers (and the important execution options they present) in a market with queuing.

Regarding a dealers' decision to jump the queue, it may be rational for a dealer to assume the dealer ahead of him will consume the bulk of liquidity at a given price. The dealer's decision is then whether to wait in the queue and risk having to trade at worse prices or to switch to another broker at possibly worse prices.³¹ Alternatively, a rational dealer may assume trading will validate prices, increasing those willing to trade. Liquidity could then be non-monotonic: rising at first as more join the process and then falling as revised trading demands are satiated. Regardless of dealers' beliefs about the shape of liquidity, the ordered probit results suggest the following: if dealers want to trade in the same direction as the responding order in progress, they expect better prices by queue-jumping rather than by waiting in a queue only to learn that limit order depth has been consumed and quotes have been revised before they can trade.

V. Conclusions

The expandable order protocol provides an interesting alternative to traditional limit order books that prioritize orders by price and time. Dealers can use expandable order strategies to discover depth quickly while mitigating their free-option and free-riding problems. We find that dealers tend to use expandable order strategies to execute large trades. They appear to use expandable order strategies more frequently during periods of greater information asymmetry, such as during overseas trading hours and for off-the-run issues. There is no evidence that for on-the-run trading during New York trading hours, dealers can interpret asymmetric depth of the limit order book as a valuable information signal and attempt to free-ride on it. The expandable order protocol, however, does not eliminate the possibility of free-riding: for on-the-run trading outside New York hours and for off-the-run trading, there is evidence that asymmetric depth may provide a valuable signal about the sign of the next trade.

³¹ We thank an anonymous referee for this insight.

We conclude that broker competition, which permits queue-jumping, provides an important execution option but also caps the value of paired negotiations. We find little evidence, however, that use of the expandable order protocol in the brokered interdealer Treasury market causes market inefficiency in the form of durable crossed spreads. We find some evidence that the expandable order protocol constrains immediacy: about 14% of overlapping transactions execute with negative price improvement for the responding order. In this very transparent market, we interpret this as evidence that sometimes dealers are unwilling to wait in queues while others trade. Ordered probit analysis of expected price changes suggests dealers should attempt to trade via an overlapping transaction only if they want to trade in the same direction as the trade already in progress. Most dealer behavior appears to be consistent with this prediction. We find 84% of overlapping transactions have the same sign as the trade already underway.

Finally, regarding the potential for the expandable order protocol in other markets, we have the following insight. Given the inherent tendency of the protocol to create queues and the apparent willingness of traders to jump queues, even at disadvantageous prices, non-competitive broker markets or electronic exchanges that use *only* the expandable order protocol may not be durable market structures. It would appear that a successful protocol involving expandable orders to encourage rapid accumulation of trading volume must also involve a multiplicity of brokers or intermittent batch-crossing sessions to limit the negative externality imposed on those waiting to trade.

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Table I
Summary Statistics for Five-Year Treasury Notes for October 1-29, 1997

Cusip 9128273	J9	G5	C4
Trading Status	On-the Run	Off-the Run	Off-the Run
Issue Date	9/30/1997	9/02/1997	7/31/1997
Maturity Date	9/30/2002	8/31/2002	7/31/2002
Interest Rate (%)	5 ⁷ / ₈	6 ¹ / ₄	6
Par Amount Issued by Treasury (Million \$)	12,807	12,732	12,231
Average Daily Volume Traded (Million \$/Day)	8,704	70	7
Average Daily Number of Transactions	1,045	16	1
Average Daily Closing Price (% of Par)	99.756	101.116	100.079
<u>Transaction Size (Million \$)</u>			
Mean	8.3	4.4	6.1
10%	1	1	1
Median	5	4	4
90%	19	10	20
<u>Ratio Takes/Hits:</u>			
Million \$ Transacted	1.07	1.05	2.17
Number of Transactions	1.06	0.97	1.40
<u>Quote Spread:</u>			
% of Time Both Bid and Ask Are Present	89.0%	67.9%	25.9%
Time-Weighted Spread Conditional on Both Bid and Ask Present (% of Par)	0.01540	0.03097	0.02862
<u>Quote Revisions:</u>			
Average Time Between Revisions (Seconds)	17.9	42.3	156.3
Average Daily Number of Bid or Ask Revisions	4,801	2,024	548
Ratio Revisions/Number of Transactions	4.6	127.3	456.7
<u>Workup Statistics</u>			
Average Daily Number of Workups	1,934.0	28.4	1.9
Median Workup Size	3	1	3
Average Time (Seconds) Between First Workup and Transaction Posting for Transactions of:			
1 Workup	13.4	15.1	26.4
2 Workups	19.7	19.0	24.5
More Than 2 Workups	29.2	20.6	29.3

Table II

Transactions by Size and Quoted Depth Immediately Prior to Transaction

				<i>Memo: Transactions with Negative Price Improvement</i>	
				Number of Transactions	Volume (M\$) of Transactions
<i>Panel A: On-the-Run Issue, New York Trading Hours</i>					
Single-Workup Transactions with:					
		Number of Transactions	Dollar Volume of Transactions, \$M		
		Column Percent	Column Percent		
Size ≤ Quoted Depth				547	1,392
Size > Quoted Depth				484	4,434
Multiple-Workup Transactions with:					
Size ≤ Quoted Depth		2,104	18,376	85	450
Size > Quoted Depth:					
& Initial Workup > Quoted Depth		437	10,288	146	3,138
& Initial Workup ≤ Quoted Depth		4,447	61,111	138	1,511
All Transactions		18,888	157,435	1,400	10,925
<i>Panel B: On-the-Run Issue, Non-New York Trading Hours</i>					
Single-Workup Transactions with:					
Size ≤ Quoted Depth		781	3,046	14	18
Size > Quoted Depth		93	972	11	77
Multiple-Workup Transactions with:					
Size ≤ Quoted Depth		102	626	-	-
Size > Quoted Depth:					
& Initial Workup > Quoted Depth		49	963	10	126
& Initial Workup ≤ Quoted Depth		977	11,030	7	59
All Transactions		2,002	16,637	42	280
<i>Panel C: Off-the-Run Issues, All Trading Hours</i>					
Single-Workup Transactions with:					
Size ≤ Quoted Depth		200	545	11	11
Size > Quoted Depth		6	33	2	15
Multiple-Workup Transactions with:					
Size ≤ Quoted Depth		10	74	1	2
Size > Quoted Depth:					
& Initial Workup > Quoted Depth		1	18	1	18
& Initial Workup ≤ Quoted Depth		124	882	1	2
All Transactions		341	1,552	16	48

Table III
Multiple- and Single-Workup Transactions by Trade Size

For the sample period October 1 through October 29, 1997, this table reports the empirical percentage frequency of transactions with multiple and single workups, conditional on trade size category. Each row is a probability vector, conditional on trade size category, summing to 100 percent. Trade size categories are small trades (1 to 4 million dollars), medium trades (5 to 9 million dollars), and large trades (10 million dollars or more).

	Multiple-Workup Transactions	Single-Workup Transactions
<i>Panel A: On-the-Run Transactions, All Trading Hours</i>		
Small Trades	19.2	80.8
Medium Trades	36.2	63.8
Large Trades	62.6	37.4
<i>Panel B: Off-the-Run Transactions, All Trading Hours</i>		
Small Trades	25.4	74.6
Medium Trades	47.4	52.6
Large Trades	83.0	17.0
<i>Panel C: On-the-Run Transactions, New York versus Non-New York Trading Hours</i>		
<i>New York Trading Hours:</i>		
Small Trades	18.0	82.0
Medium Trades	33.7	66.3
Large Trades	60.6	39.4
<i>Non-New York Trading Hours:</i>		
Small Trades	29.4	70.6
Medium Trades	62.9	37.1
Large Trades	81.0	19.0
<i>Panel D: On-the-Run Transactions, 8:30 A.M. to 8:45 A.M.</i>		
<i>Days Without 8:30 A.M. Macroeconomic Announcement:</i>		
Small Trades	21.3	78.7
Medium Trades	31.1	68.9
Large Trades	53.9	46.1
<i>Days With 8:30 A.M. Macroeconomic Announcement:</i>		
Small Trades	22.2	77.8
Medium Trades	32.1	67.9
Large Trades	62.8	37.2

Table IV
Initial Workup Size Relative to the Size of Other Workups

For the sample period October 1 through October 29, 1997, this table reports the empirical percentage frequency of multiple-workup transactions categorized by the initial workup size relative to the average size of the other workups in the transaction. Each row is a probability vector summing to 100 percent.

	Initial Workup Size > Average Size of Other Workups in Transaction	Initial Workup Size = Average Size of Other Workups in Transaction	Initial Workup Size < Average Size of Other Workups in Transaction
<i>Panel A: On-the-Run Transactions, All Trading Hours</i>			
	25.3	23.0	51.7
<i>Panel B: Off-the-Run Transactions, All Trading Hours</i>			
	11.1	28.2	60.7
<i>Panel C: On-the-Run Transactions, New York versus Non-New York Trading Hours</i>			
<i>New York Trading Hours:</i>			
	27.6	24.1	48.3
<i>Non-New York Trading Hours:</i>			
	10.9	16.4	72.7
<i>Panel D: On-the-Run Transactions, 8:30 A.M. to 8:45 A.M.</i>			
<i>Days Without 8:30 A.M. Macroeconomic Announcement:</i>			
	37.8	23.5	38.7
<i>Days With 8:30 A.M. Macroeconomic Announcement:</i>			
	28.3	23.9	47.8

Table V
Frequency of Minimum-Size Initial Workups

For the sample period October 1 through October 29, 1997, this table reports the empirical percentage frequency of transactions categorized by whether the initial workup is of minimum size (\$ 1 million). Each row is a probability vector summing to 100 percent.

	Initial Workup Size Equals \$ 1 Million	Initial Workup Size Exceeds \$ 1 Million
<i>Panel A: On-the-Run Transactions, All Trading Hours</i>		
	37.6	62.4
<i>Panel B: Off-the-Run Transactions, All Trading Hours</i>		
	61.9	38.1
<i>Panel C: On-the-Run Transactions, New York versus Non-New York Trading Hours</i>		
<i>New York Trading Hours:</i>		
	35.0	65.0
<i>Non-New York Trading Hours:</i>		
	62.0	38.0
<i>Panel D: On-the-Run Transactions, 8:30 A.M. to 8:45 A.M.</i>		
<i>Days Without 8:30 A.M. Macroeconomic Announcement:</i>		
	31.9	68.1
<i>Days With 8:30 A.M. Macroeconomic Announcement:</i>		
	37.4	62.6

Table VI
Means and Medians of Initial Workups

For the sample period October 1 through October 29, 1997, this table reports the mean and median of the initial workups of transactions in millions of dollars.

	Mean		Median
On-the-Run Transactions, All Trading Hours	4.92	} 0.000	4
Off-the-Run Transactions, All Trading Hours	2.45		1
<i>On-the-Run Transactions:</i>			
• New York Trading Hours	5.11	} 0.000	4
• Non-New York Trading Hours	3.18		1
<i>Panel D: On-the-Run Transactions, 8:30 A.M. to 8:45 A.M.</i>			
• Days Without 8:30 A.M. Macroeconomic Announcement	5.22	} 0.554	5
• Days With 8:30 A.M. Macroeconomic Announcement	4.98		4

indicates p-value for means test ($H_0: \mu_i = \mu_j$).

Table VII
Information Free-Riding and Asymmetric Quoted Depth

For the sample period October 1 through October 29, 1997, this table reports the empirical percentage frequency of transactions by trade sign, conditional on relative quoted depth immediately prior to first workup. Each row is a probability vector, conditional on relative quoted depth, summing to 100 percent. Relative quoted depth categories are ask depth > bid depth, ask depth = bid depth, and ask depth < bid depth. Unconditional probability of hit or take is shown in boldface type.

	Hit Transactions	Take Transactions
<i>Panel A: On-the-Run Transactions, All Trading Hours</i>		
Ask Depth > Bid Depth	49.6	50.4
Ask Depth = Bid Depth	47.8	52.2
Ask Depth < Bid Depth	47.6	52.4
Unconditionally	48.5	51.5
<i>Panel B: Off-the-Run Transactions, All Trading Hours</i>		
Ask Depth > Bid Depth	56.8	43.2
Ask Depth = Bid Depth	53.5	46.5
Ask Depth < Bid Depth	40.7	59.3
Unconditionally	50.1	49.9
<i>Panel C: On-the-Run Transactions, New York versus Non-New York Trading Hours</i>		
<i>New York Trading Hours:</i>		
Ask Depth > Bid Depth	49.7	50.3
Ask Depth = Bid Depth	49.2	50.8
Ask Depth < Bid Depth	48.2	51.8
Unconditionally	49.0	51.0
<i>Non-New York Trading Hours:</i>		
Ask Depth > Bid Depth	49.0	51.0
Ask Depth = Bid Depth	43.0	57.0
Ask Depth < Bid Depth	40.9	59.1
Unconditionally	44.5	55.5
<i>Panel D: On-the-Run Transactions, 8:30 A.M. to 8:45 A.M.</i>		
<i>Days Without 8:30 A.M. Macroeconomic Announcement:</i>		
Ask Depth > Bid Depth	55.9	44.1
Ask Depth = Bid Depth	52.2	47.8
Ask Depth < Bid Depth	46.7	53.3
Unconditionally	51.2	48.8
<i>Days With 8:30 A.M. Macroeconomic Announcement:</i>		
Ask Depth > Bid Depth	56.3	43.7
Ask Depth = Bid Depth	55.1	44.9
Ask Depth < Bid Depth	51.3	48.7
Unconditionally	53.8	46.2

Table VIII
Transaction Overlap (On-the-Run Issue)

For the sample period October 1-29, 1997, this table reports non-overlapping and overlapping transactions for on-the-run 5-year Treasury notes. Transactions are further broken out by whether they have price improvement, as defined in "Notes" below.

	<u>Negative Price Improvement</u>	<u>No Price Improvement</u>	<u>Positive Price Improvement</u>	<u>All</u>
<i>Non-overlapping Transactions</i>				
Transaction Volume, Millions	\$ 2,701	\$ 108,466	\$ 700	\$ 111,867
Number of Transactions	370	12,662	61	13,093
Average Transaction Size, Millions	\$ 7.30	\$ 8.57	\$ 11.48	\$ 8.54
<i>Overlapping Transactions</i>				
Transaction Volume, Millions	\$ 8,504	\$ 53,398	\$ 303	\$ 62,205
Number of Transactions	1,072	6,711	14	7,797
Average Transaction Size, Millions	\$ 7.93	\$ 7.96	\$ 21.64	\$ 7.98
<i>Non-overlapping + Overlapping</i>				
Transaction Volume, Millions	\$ 11,205	\$ 161,864	\$ 1,003	\$ 174,072
Number of Transactions	1,442	19,373	75	20,890
Average Transaction Size, Millions	\$ 7.77	\$ 8.36	\$ 13.37	\$ 8.33

Notes:

- 1.) "Negative price improvement" is defined as transaction price > best ask for take transactions and transaction price < best bid for hit transactions.
- 2.) "No price improvement" is defined as transaction price = best ask for take transactions and transaction price = best bid for hit transactions.
- 3.) "Positive price improvement" is defined as transaction price < best ask for take transactions and transaction price > best bid for hit transactions.
- 4.) "*" indicates we reject $H_0: \mu_i = \mu_j$ at the 5% significance level. "=" indicates we cannot reject.

Table IX

Maximum likelihood estimates for ordered probit model of price changes for on-the-run five-year Treasury notes traded October 1-29, 1997.

P-value, based on 2-tailed test of null hypothesis that parameter is zero, is shown in italics to right of estimate.

Panel A: New York hours		Model: Hits & Takes		Model: Takes Only		Model: Hits Only	
Observations of Price Change of:							
$s_1 \leq -3/128ths \text{ of } 1\% \text{ of face}$		244		31		213	
$s_2 = -2/128ths \text{ of } 1\% \text{ of face}$		1,043		145		898	
$s_3 = -1/128ths \text{ of } 1\% \text{ of face}$		2,418		566		1,852	
$s_4 = 0.$		11,393		5,847		5,546	
$s_5 = 1/128ths \text{ of } 1\% \text{ of face}$		2,472		1,933		539	
$s_6 = 2/128ths \text{ of } 1\% \text{ of face}$		1,090		929		161	
$s_7 \geq 3/128ths \text{ of } 1\% \text{ of face}$		219		185		34	
Parameter							
α_1		-2.8089	0.0000	-3.1422	0.0000	-2.3511	0
α_2		-1.8606	0.0000	-2.4177	0.0000	-1.3859	0
α_3		-1.0534	0.0000	-1.6279	0.0000	-0.5558	0
α_4		1.0464	0.0000	0.5972	0.0000	1.6090	0
α_5		1.8651	0.0000	1.4517	0.0000	2.3392	0
α_6		2.9020	0.0000	2.5071	0.0000	3.1030	0
$\beta_1:$	Bid_k	-0.00346	0.0000	-0.00397	0.0000	-0.10980	0
$\beta_2:$	Ask_k	0.00314	0.0000	0.01500	0.0000	0.00239	0
$\beta_3:$	$(1-IND_k)(Z_{k-1})$	-13.81223	0.0000	-30.71928	0.0000	-8.87445	0
$\beta_4:$	$(1-IND_k)(IBS_{k-1})$	0.37029	0.0000	0.23133	0.0000	0.14438	0
$\beta_5:$	$(1-IND_k)(V_{k-1}*IBS_{k-1})$	-0.00009	0.9474	-0.00138	0.4504	0.00138	0
$\beta_6:$	$(1-IND_k)(N_{k-1}*IBS_{k-1})$	-0.01184	0.0986	-0.00452	0.6122	0.00425	0
$\beta_7:$	$(IND_k)(Z_{k-1})$	-23.62656	0.0000	-26.30246	0.0000	-33.05127	0
$\beta_8:$	$(IND_k)(IBS_{k-1})$	0.15248	0.0000	-0.26425	0.0000	-0.24700	0
$\beta_9:$	$(IND_k)(V_{k-1}*IBS_{k-1})$	-0.00398	0.0139	-0.00723	0.0021	-0.00379	0
$\beta_{10}:$	$(IND_k)(N_{k-1}*IBS_{k-1})$	0.04230	0.0000	0.07670	0.0000	0.06573	0
$\gamma_1:$	Δt_k	0.13034	0.0000	0.09137	0.0000	0.09204	0

Table IX (continued)

Maximum likelihood estimates for ordered probit model of price changes for on-the-run five-year Treasury notes traded October 1-29, 1997.
P-value, based on 2-tailed test of null hypothesis that parameter is zero, is shown in italics to right of estimate.

Panel B: Non-New York hours	Model:		Model:	
	Hits & Takes	Takes Only	Hits Only	
Observations of Price Change of:				
$s_1 \leq -3/128\text{ths of } 1\% \text{ of face}$	71	14	57	
$s_2 = -2/128\text{ths of } 1\% \text{ of face}$	128	25	103	
$s_3 = -1/128\text{ths of } 1\% \text{ of face}$	216	70	146	
$s_4 = 0.$	1,104	613	491	
$s_5 = 1/128\text{ths of } 1\% \text{ of face}$	267	201	66	
$s_6 = 2/128\text{ths of } 1\% \text{ of face}$	141	127	14	
$s_7 \geq 3/128\text{ths of } 1\% \text{ of face}$	73	61	12	
Parameter				
α_1	-2.7505	0.0000	-2.7460	0.0000
α_2	-1.8799	0.0000	-2.1771	0.0000
α_3	-1.1490	0.0000	-1.4981	0.0000
α_4	1.0590	0.0000	0.5678	0.0000
α_5	1.8697	0.0000	1.2724	0.0000
α_6	2.7559	0.0000	2.0941	0.0000
$\beta_1: Bid_k$	-0.00074	0.8636	-0.00021	0.9604
$\beta_2: Ask_k$	-0.00407	0.3676	-0.00072	0.9134
$\beta_3: (1-IND_k)(Z_{k-1})$	-2.71992	0.3235	-7.27542	0.0356
$\beta_4: (1-IND_k)(IBS_{k-1})$	0.64362	0.0000	0.47192	0.0000
$\beta_5: (1-IND_k)(V_{k-1} * IBS_{k-1})$	-0.01093	0.0057	-0.01132	0.0159
$\beta_6: (1-IND_k)(N_{k-1} * IBS_{k-1})$	-0.01439	0.5579	0.00319	0.9068
$\beta_7: (IND_k)(Z_{k-1})$	-2.82796	0.6488	-14.57825	0.0740
$\beta_8: (IND_k)(IBS_{k-1})$	-0.05367	0.6570	-0.45166	0.0089
$\beta_9: (IND_k)(V_{k-1} * IBS_{k-1})$	0.01183	0.0891	0.01526	0.1415
$\beta_{10}: (IND_k)(N_{k-1} * IBS_{k-1})$	0.03880	0.5891	0.00575	0.9589
$\gamma_1: \Delta t_k$	0.06154	0.0000	0.03288	0.0000
			0.06694	0.0000
				44

Table X

Expected Price Changes (Previous Transaction Is a "Take", New York Hours *)

<i>Panel A: Expected price change for responding take order ($E[Z_k take]$), \$ per M \$ traded</i>				
Z_{k-1}	N_{k-1}	V_{k-1}	$E[Z_k take] : Transaction_k$ <i>is overlapping</i>	$E[Z_k take] : Transaction_k$ <i>is not overlapping</i>
0	1	1	19.96	42.26
0	2	10	20.54	42.26
$+ \frac{1}{128}$	1	1	-1.46	19.00
$+ \frac{1}{128}$	2	10	-0.90	19.00
$- \frac{1}{128}$	1	1	43.35	68.78
$- \frac{1}{128}$	2	10	44.00	68.78
<i>Panel B: Expected price change for responding hit order ($E[Z_k hit]$), \$ per M \$ traded</i>				
Z_{k-1}	N_{k-1}	V_{k-1}	$E[Z_k hit] : Transaction_k$ <i>is overlapping</i>	$E[Z_k hit] : Transaction_k$ <i>is not overlapping</i>
0	1	1	-41.00	-22.90
0	2	10	-39.19	-22.90
$+ \frac{1}{128}$	1	1	-60.73	-40.89
$+ \frac{1}{128}$	2	10	-58.75	-40.89
$- \frac{1}{128}$	1	1	-23.00	-6.08
$- \frac{1}{128}$	2	10	-21.33	-6.08
<i>Panel C: Expected price change ($E[Z_k]$), \$ per M \$ traded</i>				
Z_{k-1}	N_{k-1}	V_{k-1}	$E[Z_k] : Transaction_k$ <i>is overlapping</i>	$E[Z_k] : Transaction_k$ <i>is not overlapping</i>
0	1	1	9.93	19.58
0	2	10	10.28	19.58
$+ \frac{1}{128}$	1	1	-5.60	3.93
$+ \frac{1}{128}$	2	10	-5.25	3.93
$- \frac{1}{128}$	1	1	25.77	35.84
$- \frac{1}{128}$	2	10	26.13	35.84

* Note: bid size and ask size = \$ 10 million, Δt_k is 20 seconds.

Figure 1

Example of Brokered Interdealer Treasury Note Trading

Note: Sequence of events at Time t is that actions described at Time t occur and Broker and GovPX screens update at Time t to reflect those actions.

	Broker X's Screen		Broker Y's Screen		GovPX Screen *	
	<u>Bid</u>	<u>Ask</u>	<u>Bid</u>	<u>Ask</u>	<u>Bid</u>	<u>Ask</u>
<u>Time 0:</u> No limit orders placed.	-	-	-	-	-	-
<u>Time 1:</u> Dealer A places BUY limit order with Broker X for \$ 1 M at $100^1/_{64}$.	$100^1/_{64} \times 1$	-	-	-	$100^1/_{64} \times 1$	-
<u>Time 2:</u> Dealer B places SELL limit order with Broker X for \$ 1 M at $100^3/_{64}$.	$100^1/_{64} \times 1$	$100^3/_{64} \times 1$	-	-	$100^1/_{64} \times 1$	$100^3/_{64} \times 1$
<u>Time 3:</u> Dealer C places BUY limit order with Broker X for \$ 1 M at $100^1/_{64}$.	$100^1/_{64} \times 2$	$100^3/_{64} \times 1$	-	-	$100^1/_{64} \times 2$	$100^3/_{64} \times 1$
<u>Time 4:</u> Dealer D tells Broker X to hit \$ 5 M at $100^1/_{64}$. Broker X tells Dealer D he is done on \$ 1 M. Broker X tells Dealer A he bought \$ 1 M at $100^1/_{64}$ and asks whether he has more to buy. <i>{Workup field flashes price and \$ volume.}</i>	$100^1/_{64} \times 2$	$100^3/_{64} \times 1$ $100^1/_{64} \times 1$	-	-	$100^1/_{64} \times 2$	$100^3/_{64} \times 1$ $100^1/_{64} \times 1$
<u>Time 5:</u> Dealer A says he will buy \$ 2 M more at $100^1/_{64}$. Broker X tells Dealer D he sold another \$ 2 M. <i>{Workup field flashes price and total \$ volume.}</i>	$100^1/_{64} \times 2$	$100^3/_{64} \times 1$ $100^1/_{64} \times 3$	-	-	$100^1/_{64} \times 2$	$100^3/_{64} \times 1$ $100^1/_{64} \times 3$
<u>Time 6:</u> Broker X asks Dealer A whether he has more to buy at $100^1/_{64}$. <i>{Workup field flashes price and total \$ volume.}</i>	$100^1/_{64} \times 2$	$100^3/_{64} \times 1$ $100^1/_{64} \times 3$	-	-	$100^1/_{64} \times 2$	$100^3/_{64} \times 1$ $100^1/_{64} \times 3$

* GovPX screen is described in Section II.

Figure 1 (continued)

Example of Brokered Interdealer Treasury Note Trading

	Broker X's Screen		Broker Y's Screen		GovPX Screen *	
	Bid	Ask	Bid	Ask	Bid	Ask
<u>Time 7:</u> Dealer B places SELL limit order with Broker Y for \$ 1 M at $100\frac{1}{64}$. <i>{Workup field flashes price and total \$ volume.}</i>	$100\frac{1}{64} \times 2$	$100\frac{3}{64} \times 1$ $100\frac{1}{64} \times 3$	-	$100\frac{1}{64} \times 1$	$100\frac{1}{64} \times 2$ $100\frac{1}{64} \times 3$	$100\frac{1}{64} \times 1$
<u>Time 8:</u> Dealer A tells Broker X he has no more to buy at $100\frac{1}{64}$. <i>{Workup field flashes price and total \$ volume.}</i>	$100\frac{1}{64} \times 1$	$100\frac{3}{64} \times 1$ $100\frac{1}{64} \times 3$	-	$100\frac{1}{64} \times 1$	$100\frac{1}{64} \times 1$ $100\frac{1}{64} \times 3$	$100\frac{1}{64} \times 1$
<u>Time 9:</u> Broker X tells Dealer C he bought \$ 1 M at $100\frac{1}{64}$ and asks whether he has more to buy. Broker X tells Dealer D he sold another \$ 1 M. <i>{Workup field flashes price and total \$ volume.}</i>	$100\frac{1}{64} \times 1$	$100\frac{3}{64} \times 1$ $100\frac{1}{64} \times 4$	-	$100\frac{1}{64} \times 1$	$100\frac{1}{64} \times 1$ $100\frac{1}{64} \times 4$	$100\frac{1}{64} \times 1$
<u>Time 10:</u> Dealer C tells Broker X he wants no more at $100\frac{1}{64}$. Broker X tells Dealer D he has no more bid at $100\frac{1}{64}$. Broker X clears workups from screen and adds total of \$ 4 M to transaction volume field (not shown here).	-	$100\frac{3}{64} \times 1$	-	$100\frac{1}{64} \times 1$	-	$100\frac{1}{64} \times 1$
<u>Time 11:</u> Dealer A places BUY limit order with Broker X for \$ 1 M at 100.	100×1	$100\frac{3}{64} \times 1$	-	$100\frac{1}{64} \times 1$	100×1	$100\frac{1}{64} \times 1$
<u>Time 12:</u> Broker X asks Dealer C whether he wants to hit the bid at 100.	100×1	$100\frac{3}{64} \times 1$	-	$100\frac{1}{64} \times 1$	100×1	$100\frac{1}{64} \times 1$

* GovPX screen is described in Section II.

Figure2 DailyClosingPrices

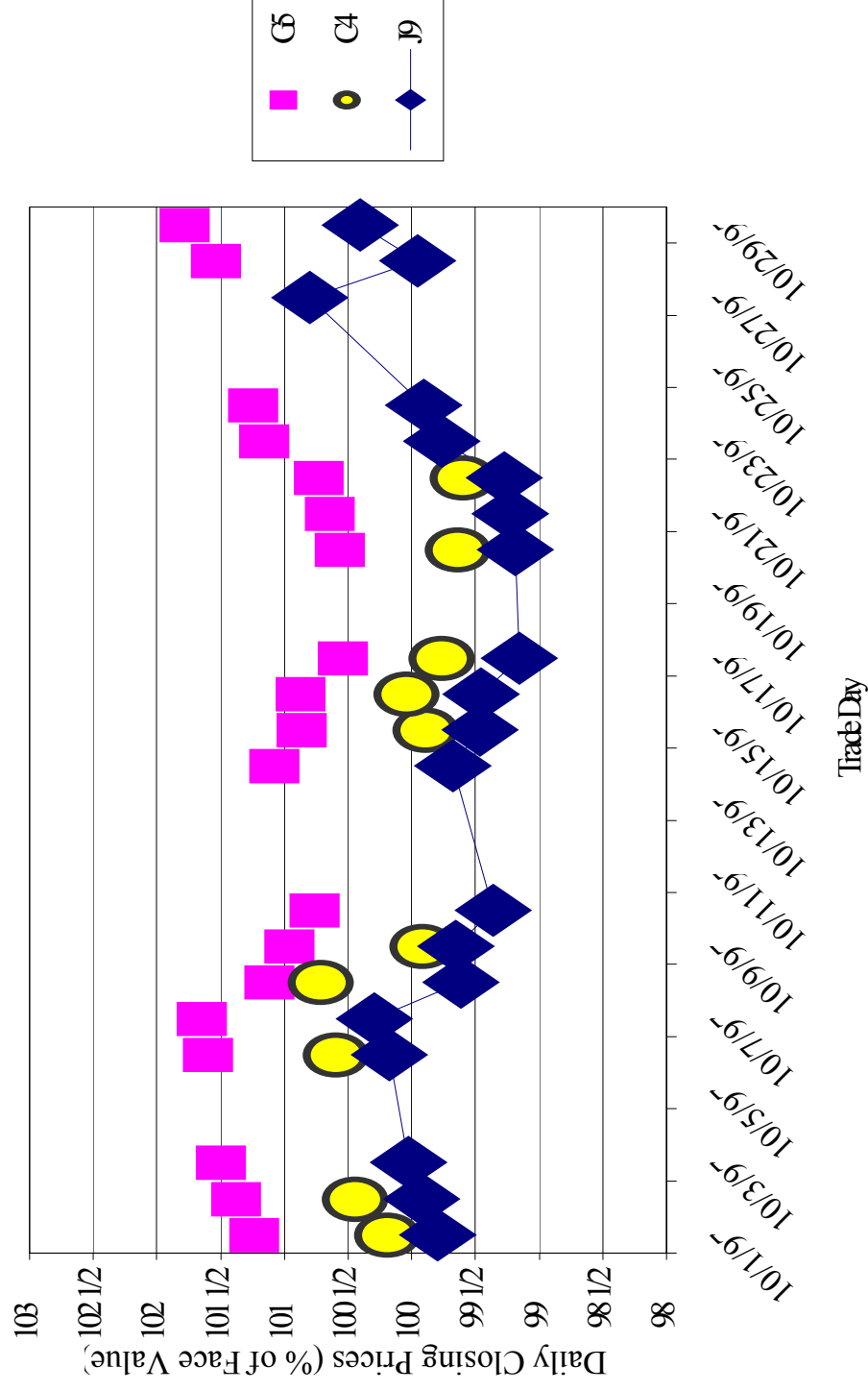


Figure 3
On-the-Run Five-Year Treasury Notes
Daily Transaction Volumes

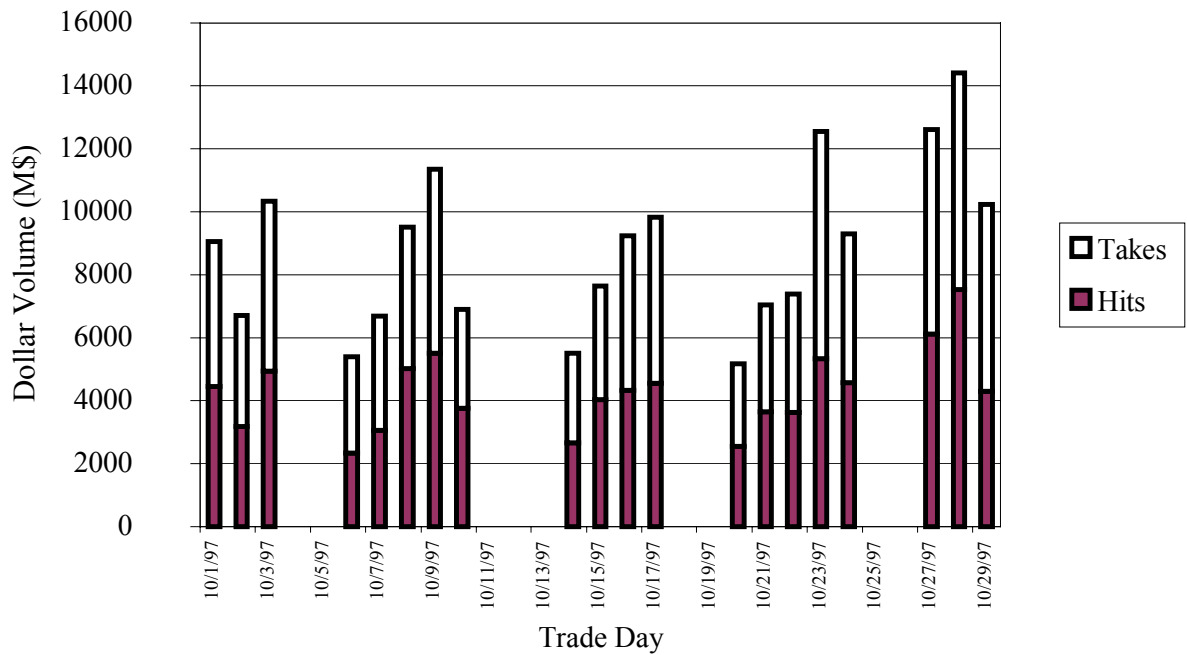
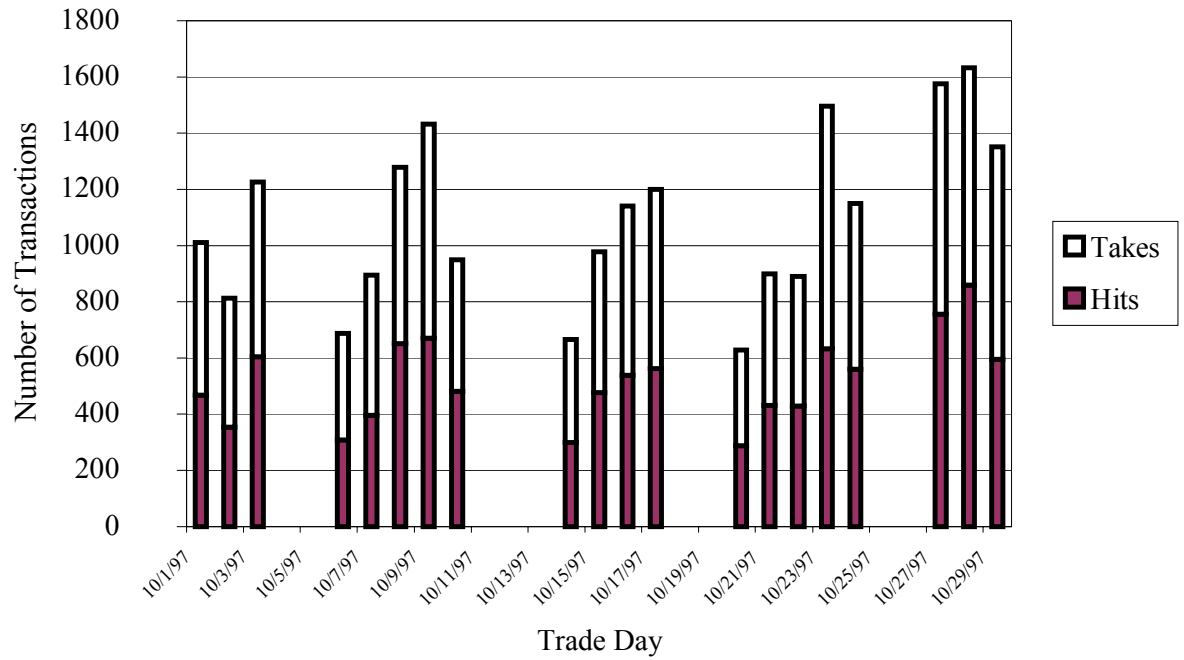


Figure 4
Five-Year Treasury Notes
Price Change Between Transactions (All Hours)

